

When Cross-View Helps Most: Conflict-Aware Disaster Triage Across Spatial Observation Regimes [experiments]

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Abstract

Cross-view disaster assessment combines ground-level and overhead imagery to infer building damage, but aggregate F1 hides the examples where the two views disagree. This paper introduces Conflict-Aware Evaluation (CAE), a reporting protocol that measures overall F1, conflict-subset F1, cross-view gain over the stronger single-view baseline, and the direction in which one view corrects the other. We apply CAE to an Eaton/Altadena wildfire dataset with property-centered ground imagery and to the Hurricane Ian CVDIsaster benchmark with panoramic ground imagery. Results differ by observation regime. On the wildfire sensitivity split, cross-view fusion improves conflict F1 by $\Delta_{\text{view}} = +0.0388$, and a label-independence permutation test rejects the null at $p < 10^{-4}$. On the hurricane endpoint task, the picture inverts once spatial leakage is removed: the same task shows a $+0.0805$ conflict gain under an object-leaky split but a slightly negative gain ($\Delta_{\text{view}} = -0.0153$, within overlapping bootstrap intervals) under an object-grouped clean split, even though aggregate F1 makes cross-view and street-only models look nearly tied. Directional and alignment diagnostics tie this to how the ground view sees the target: wildfire conflict images contain roughly seventeen times more building pixels than hurricane conflict images, and fusion loses value when the ground view becomes panoramic context rather than a property-centered observation. CAE therefore turns paired-view triage into a stress test for when fusion helps, fails, or should defer.

Keywords

disaster assessment, cross-view, multimodal fusion, GeoAI, conflict-aware evaluation

1 Introduction

Post-disaster building assessment increasingly uses several visual viewpoints. Overhead imagery gives broad coverage, while ground imagery can expose facade condition, debris, and local structural cues. Cross-view models seek to combine these signals, building on a long line of cross-view geolocalization and matching work that aligns ground and aerial observations of the same place [6]. Yet average test scores hide the cases that matter most for deployment: the buildings where the views disagree.

This paper treats paired-view triage as conflict resolution. Each example has a ground image, an overhead patch, and a binary damage label. We ask whether fusion improves the subset where the single-view predictors disagree, rather than whether it only

improves average F1. This framing matters because a property inspection photograph and a panoramic street scene are both ground views, yet they observe the labeled building in very different ways.

The work builds on overhead damage benchmarks [2], fusion models for disaster assessment [7], ground-view damage studies [8, 10, 11], and the paired Hurricane Ian CVDIsaster benchmark [4]. CAE is related to selective prediction [1] because both settings identify cases where an automatic decision may need extra scrutiny, but CAE does not estimate risk-coverage curves or abstention. Its reporting unit is spatial: the disagreement subset, the single-view correction direction, and the gain or loss from fusion under a given observation regime.

We make three contributions. (i) We define CAE, a compact protocol that reports conflict-subset F1, the directional arbitration balance, and the cross-view conflict gain Δ_{view} alongside aggregate F1. (ii) Using CAE across two disasters and four splits, we show that the value of fusion tracks the *spatial observation regime*—how a ground view aligns with the labeled building—more closely than it tracks aggregate F1. (iii) We show that the most widely reported reading of cross-view benefit can be an artifact of spatial leakage: on a single hurricane endpoint task, controlling object-level leakage flips the conflict gain from clearly positive to slightly negative, turning a “cross-view helps” result into a deployment boundary that CAE makes visible.

2 Data and Protocol

2.1 Paired-View Triage Settings

Let $\mathcal{D} = \{(x_i^{(s)}, x_i^{(r)}, y_i)\}_{i=1}^N$ be a paired disaster triage dataset with a ground image, an overhead patch, and a binary label. We compare street-only (S), remote-only (R), and cross-view (CV) predictors. CV uses late fusion of ResNet18 features [3]: both embeddings, their absolute difference, and their elementwise product. The three predictors share the same backbone and training budget so that CAE compares evidence access, not model capacity.

Table 1 summarizes the four evaluation settings. The wildfire setting uses 19,720 usable Eaton/Altadena paired records derived from property-level inspection photographs and matched overhead crops. The positive class is Affected + Minor + Major + Destroyed. The hurricane setting uses the Hurricane Ian CVDIsaster benchmark [4], with 4,121 paired samples across minor, moderate, and severe damage. We evaluate the endpoint task (minor versus severe) under both a leaky and a leakage-controlled split, and a broader moderate-plus-severe task. We use these settings to contrast property-centric and panoramic observation regimes.

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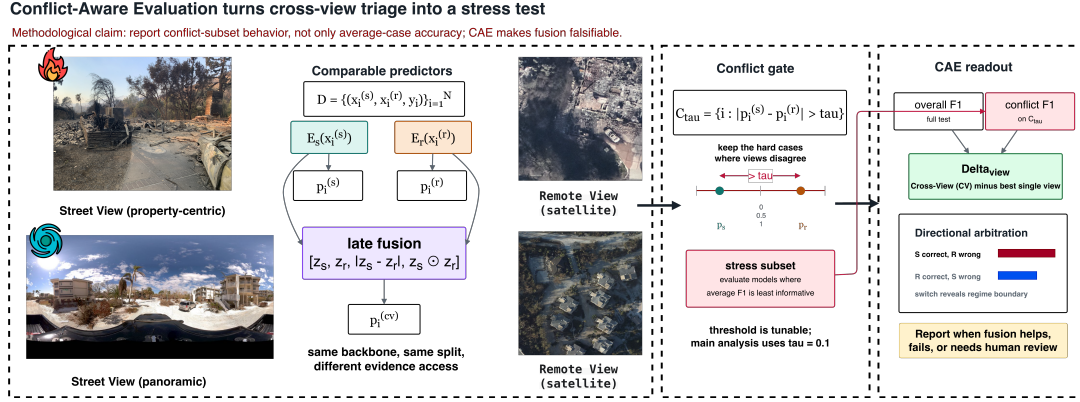


Figure 1: Conflict-Aware Evaluation (CAE) protocol. CAE isolates examples where street-only and remote-only predictors disagree, then reports whether cross-view fusion improves conflict-subset F1 over the stronger single-view baseline. Directional arbitration links model behavior to spatial observation regime.

Table 1: Evaluation settings, sample counts, and leakage controls. The two hurricane endpoint rows are the *same* minor-versus-severe task and differ only in whether object-level leakage is removed.

Setting	Positive class	Ground regime	N	Role
WF sens.	affected+	property	19,720	main CAE
H end. (leaky)	severe	panoramic	2,583	diagnostic
H mod.+sev.	moderate+	panoramic	4,121	sensitivity
H end. (clean)	severe	panoramic	2,583	leakage control

Wildfire data statement. The wildfire labels are official CAL FIRE Damage Inspection (DINS) ratings for the 2025 Eaton fire in Altadena, CA, published through the CAL FIRE incident portal.¹ Trained CAL FIRE inspectors assign each structure one of six categories; by ground-photo count these are Destroyed (9,490), Major (155), Minor (312), Affected (1,809), No Damage (7,981), and Inaccessible (33), spanning 18,388 usable structures. We map the four damage classes to the positive label and No Damage to the negative label, and exclude Inaccessible. This leaves 19,747 candidate photos (11,766 positive, 7,981 No Damage); pairing each with its overhead crop and dropping the 27 photos without a valid pairing yields the 19,720 paired records used here. Because these are authoritative field assessments rather than crowd annotations, no inter-annotator agreement statistic applies; quality control follows the standardized DINS protocol. Every record retains an object identifier, parcel coordinates, the crop box, the overhead tile ID, and coverage flags, so the benchmark supports object-level grouping and tile-level aggregation. We use an object-grouped 80/10/10 split that keeps all photos of a structure in one partition, and release split manifests and a data card rather than raw imagery so the grouped split stays auditable.

2.2 Conflict-Aware Evaluation

For each example, let $p_i^{(s)}$ and $p_i^{(r)}$ denote the positive-class probabilities predicted by the street-only and remote-only models. CAE

defines the conflict subset

$$C_{\tau} = \{i : |p_i^{(s)} - p_i^{(r)}| > \tau\}, \quad (1)$$

with $\tau = 0.1$ in the main analysis. For each model m , CAE reports standard overall F1 and conflict F1 on C_{τ} . It then computes the cross-view conflict gain

$$\Delta_{view} = F1(\text{crossview}; C_{\tau}) - \max\{F1(\text{street}; C_{\tau}), F1(\text{remote}; C_{\tau})\}. \quad (2)$$

The compact CAE report for each model is therefore

$$r_{\tau}(m) = (F1(m; \mathcal{D}), F1(m; C_{\tau}), |C_{\tau}|/N). \quad (3)$$

We also decompose conflicts by correction direction. Let $\hat{y}_i^{(s)} = \mathbf{1}\{p_i^{(s)} \geq 0.5\}$ and define $\hat{y}_i^{(r)}$ analogously. With $c_i^{(v)} = \mathbf{1}\{\hat{y}_i^{(v)} = y_i\}$, the two arbitration rates are

$$\rho_{S \rightarrow O} = \frac{1}{|C_{\tau}|} \sum_{i \in C_{\tau}} c_i^{(s)} (1 - c_i^{(r)}), \quad (4)$$

$$\rho_{O \rightarrow S} = \frac{1}{|C_{\tau}|} \sum_{i \in C_{\tau}} c_i^{(r)} (1 - c_i^{(s)}). \quad (5)$$

$S \rightarrow O$ denotes street-correct and remote-wrong cases, and $O \rightarrow S$ denotes the reverse.

Leakage control. The original hurricane endpoint split allowed overlapping object identifiers across train and test, because a single panoramic image can be tied to multiple labeled footprints. Without grouping, near-duplicate views of the same object can appear on both sides of the split and inflate any model that effectively memorizes a location, an effect long documented for spatially and hierarchically structured data [5]. We therefore re-split with object-level grouping so that no identifier appears in both partitions. The grouped-clean endpoint split is our conservative control, and we keep the leaky split only as a contrast that isolates the effect of leakage on the conflict gain.

2.3 Implementation Details

Images are resized to 224 by 224 pixels and normalized for ImageNet-pretrained ResNet18 encoders. Each view embedding is projected to 256 dimensions. Models use AdamW with learning rate 10^{-4} for 10

¹<https://www.fire.ca.gov/incidents/>

Table 2: Conflict-Aware Evaluation summary. Conflict F1 uses C_τ with $\tau = 0.1$. WF is wildfire; H is hurricane. Gain (Δ_{view}) is reported only for cross-view rows. The two “H end.” blocks are one task under a leaky versus a leakage-controlled (clean) split.

Setting	M	All	C-F1	Gain	$ C_\tau $
WF sens.	S	0.9392	0.6230	–	435
WF sens.	R	0.9238	0.5237	–	435
WF sens.	CV	0.9473	0.6618	+0.0388	435
H end. (leaky)	S	0.9055	0.6842	–	32
H end. (leaky)	R	0.8969	0.5806	–	32
H end. (leaky)	CV	0.9192	0.7647	+0.0805	32
H mod.+sev.	S	0.8738	0.7324	–	128
H mod.+sev.	R	0.8873	0.7742	–	128
H mod.+sev.	CV	0.8915	0.7867	+0.0125	128
H end. (clean)	S	0.9016	0.8101	–	80
H end. (clean)	R	0.8385	0.6526	–	80
H end. (clean)	CV	0.9008	0.7949	-0.0153	80

epochs. Grouped splits use an 80/10/10 train/validation/test design with approximate label stratification, and the best-validation-F1 checkpoint is evaluated on the held-out test split. Stability checks use seeds 42, 123, and 456. Cross-view validation F1 is 0.9689 ± 0.0011 on wildfire and 0.9455 ± 0.0063 on hurricane; street-only baselines are close but lower (0.9657 ± 0.0001 and 0.9433 ± 0.0012).

Backbone checks rule out a small-CNN artifact. On cross-view validation, ResNet18/ResNet50/DINOv2/CLIP obtain 0.9676/0.9693/0.9378/0.7502 on wildfire and 0.9694/0.9470/0.7975/0.6437 on hurricane. The supervised CNN features remain strongest, suggesting that generic large-scale pretraining transfers imperfectly to this domain-specific binary damage task.

3 Results

Table 2 is the CAE scorecard. Average F1 alone would make CV look mostly helpful or tied. Conflict F1 changes the reading. Wildfire shows a clear positive conflict gain ($\Delta_{\text{view}} = +0.0388$), and CV is the strongest configuration both overall and on C_τ .

Leakage flips the hurricane verdict. The two hurricane endpoint rows are the same minor-versus-severe task; they differ only in whether object-level leakage is removed. Under the leaky split, CV looks helpful on conflicts ($\Delta_{\text{view}} = +0.0805$). After object grouping, the gain flips to $\Delta_{\text{view}} = -0.0153$ while overall F1 changes little and street-only slightly edges CV (0.9016 versus 0.9008). The flip isolates spatial leakage, not view regime, as the source of the original positive endpoint gain—exactly the kind of boundary that aggregate F1 conceals and CAE surfaces.

Uncertainty. The hurricane conflict subsets are small ($|C_\tau| = 32$ and 80 for the endpoint settings), so conflict F1 there is estimated with high variance. In our released bootstrap analyses, cross-view conflict-F1 intervals span roughly 0.19 in width on wildfire and exceed 0.35 on the hurricane subsets, and the single-view and cross-view intervals overlap on the hurricane settings. We therefore read the negative hurricane gain as a sign-level boundary signal to replicate, not a high-precision effect estimate, while the wildfire gain rests on a much larger subset ($|C_\tau| = 435$).

Directional arbitration. Figure 2 explains why the mixed gains are useful rather than inconvenient. In wildfire, $S \rightarrow O$ conflicts are

more common than $O \rightarrow S$ conflicts (0.271 versus 0.189), so property-centered ground imagery is the stronger arbitrator. In the broader hurricane split the ordering reverses (0.109 versus 0.133), consistent with panoramic views that contain more scene context and weaker target alignment. The endpoint hurricane split is closer to balanced (0.375 versus 0.344), reinforcing that panoramic regimes are split-sensitive.

Significance and threshold sensitivity. A label-independence permutation test on the conflict subset supports the wildfire finding: with $\tau = 0.1$ and $|C_\tau| = 435$, cross-view conflict accuracy is 0.7885 and the permutation null is rejected at $p < 10^{-4}$. On the hurricane moderate-plus-severe subset ($|C_\tau| = 128$), cross-view conflict accuracy is 0.7500 with $p < 10^{-4}$ as well, but the two small endpoint subsets ($|C_\tau| = 32, 80$) are underpowered for a strong significance claim and we do not make one. Ordering is also stable across thresholds: at $\tau = \{0.1, 0.3, 0.5\}$, CV remains strongest on wildfire conflict accuracy (0.8349, 0.7821, 0.7538). At $\tau = 0.1$, wildfire conflict accuracy is 0.7885 for CV, 0.7356 for S, and 0.6529 for R. As τ increases, only stronger single-view contradictions remain, so accuracy can drop while the wildfire ordering persists.

4 Discussion and Conclusion

The key result in Table 2 is the negative grouped-clean hurricane gain: once object-level leakage is controlled, more visual evidence does not always help. CAE thus exposes a deployment boundary, not only a performance gain. The conflict subset is a second readout, not a replacement for the test set: its gate uses only single-view probabilities, so it can be computed before trusting fusion, and the directional counts identify which view supplies the missing cue. A fused model can therefore be useful for broad screening yet unsafe for escalation if it loses to a single view on C_τ .

Spatial observation regime makes this boundary plausible. In wildfire records the photographer aims at the labeled structure, so the ground image carries direct facade evidence; in hurricane records the panoramic image captures a street segment, and the target can be small, peripheral, or occluded. A frozen building-segmentation proxy makes this concrete: on conflict images the mean building-pixel ratio is 0.268 for wildfire versus 0.015 for hurricane (about $17\times$ higher), the center-region ratio is 0.410 versus 0.027, and the building centroid is more central in wildfire (0.296 versus 0.397). The mechanism is *how each view observes the target*, not “street versus satellite.” The proxy separates the regimes but shows no strong per-sample correlation with fusion success (Figure 2c), so we read alignment as a regime-level, not per-image, explanation; the tile-level maps are likewise a descriptive audit and warrant a spatial-autocorrelation summary (e.g. Moran’s I) before any tile-level claim. Operationally, agreement cases use the agreed decision while disagreement cases route to CV or human review, and a negative Δ_{view} signals deferring to the dominant view.

Several limitations bound these claims: the two settings differ in hazard, geography, protocol, and labels, so regime is an explanatory lens, not causal proof; the alignment proxies are diagnostic, not field-verified geometry; the fusion model is simple by design; a non-learned aggregation baseline (averaging or majority vote) should be added; and object grouping matters as much as architecture [5]. A stronger benchmark would collect multiple ground-view regimes

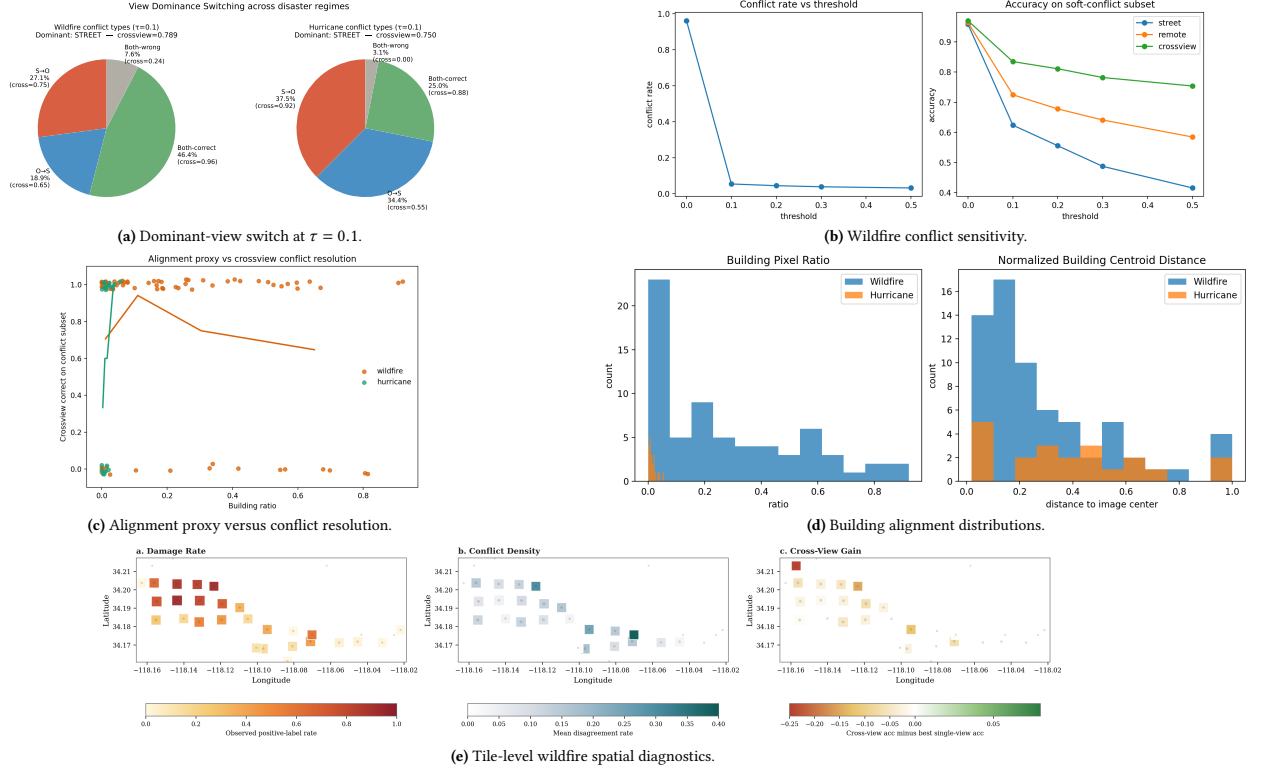


Figure 2: Result diagnostics. (a) Directional conflicts show the dominant-view switch. (b) Wildfire CV remains strongest across thresholds. (c–d) Alignment diagnostics support the regime interpretation at the dataset level. (e) Tile-level maps localize where fusion helps or fails.

for the *same* buildings, possibly with synthesized cross-view imagery [9]. In summary, CAE reframes paired-view triage as conflict resolution and reports, per view and for the fused model, the tuple $r_\tau(m)$, Δ_{view} , the conflict count $|C_\tau|$, and the $S \rightarrow O$ versus $O \rightarrow S$ balance. It shows a positive wildfire gain, a hurricane gain that flips negative once leakage is removed, and a dominant-view switch tied to regime. Benchmarks should thus report conflict-subset behavior, directional arbitration, and threshold sensitivity under grouped splits; a natural next step is conflict-aware training that reweights C_τ , pending comparison against standard focal and class-balanced losses.

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